Gayathri Krishnamoorthy

Electrical Engineer| Computer Scientist

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Power Systems Engineer with 3+ years of experience as a researcher at Washington State University and Internship experience at Pacific Northwest National Laboratory (PNNL). Worked on multiple projects developing software tools to efficiently integrate renewable resources into power grid models. I am outgoing, creative, and detail-oriented with strong engineering basics and communication skills. I am effective both solo or on a team. In my spare time I practice yoga, meditation, and painting. I am also passionate about the brain sciences behind mindfulness and keeping up with recent developments in AI.

Education

Present – August 2018 | **Ph.D. in Electrical Engineering , Washington State University, Pullman, WA**

- Research : Transmission Distribution Co-simulation : Utility of Distribution Connected Renewable Resources in Providing Transmission System Support
- Major : Power Systems
- Minor : Computer Science
- Advisor : Dr. Anamika Dubey

May 2018 – Aug 2016 | **M.S. in Electrical Engineering, Washington State University, Pullman, WA**

- Research : An Iterative Co-simulation Framework for the Integrated Transmission and Distribution System Analysis
- Major : Power Systems
- Minor : Computer Engineering
- Advisor : Dr. Anamika Dubey

May 2016 – Aug 2012 | **B.E. in Electronics and Communications Engineering, Anna University, Chennai, India**

- Title : Smart Home Design with Voice Recognition in a Single Controller (Arduino Uno) Environment
- Major : Embedded Systems
- Minor : Communications
- Advisor : Dr. Suresh T

Professional Experience

Present – August 2016 | **Graduate Research Assistant, Washington State University, Pullman, WA**

- Designed and developed a transmission and distribution (T&D) co-simulation platform to understand the impacts of increasing renewable penetrations on power system operations and evaluated various power quality issues.
- Developed analytical models to demonstrate the convergence of T&D co simulation platform.
- Developed a hybrid T&D co-simulation platform that assists in secondary frequency regulation services to the power grid using distribution connected batteries.
- Developed an optimization algorithm in a decentralized distribution market operation scenario to optimize the proposed market bids without violating system operating constraints.

MATLAB | OpenDSS | GridPV | Python | Power World

May 2019
Jan 2019

Graduate Teaching Assistant, Washington State University, Pullman, WA

Course : Professional Skills in Computing and Engineering

Tasks :

- Engaged a class of 108 undergraduate electrical and computer engineering students
- Taught the class to use Microsoft project professional software
- Detailed lectures on professional writing using LaTex.

Microsoft Project Professional | LaTex

Dec 2018
May 2018

Graduate Research Intern, Pacific Northwest National Laboratory, Seattle, WA

- Assisted in development of a transactive energy software agent, TESP, that optimizes the dynamics of distributed energy resources in a household and bids the excess energy in the energy market to provide monetary benefits to the costumers using electric vehicles, batteries, and PVs.
- Performed scalability testing for the open source co-simulation platform (HELICS) developed at PNNL in multiple programming language environments.
- Performed testing of the transactive energy simulation platform with HELICS that integrated multiple power system market components on a Linux environment
- Developed a C++ wrapper for MATPOWER software
- Developed a Docker Engine for the FNCS co-simulation software platform.

C++ | Python | MATLAB | MATPOWER | GridLAB-D | OpenDSS | Docker | EnergyPlus | FNCS | TESP | HELICS

Dec 2014

In-Plant Training, HCL Technologies, Chennai, India

- Developed fundamental principles and knowledge of recent advancements in Embedded Systems, Cloud Computing, Internet of Things, Android, and .Net software platforms.
- Attended robotics and embedded systems workshop on making of legged robots and successfully completed the project.
- Developed a prototype racing car that used Freescale controllers to automatically recognize the specially designed racetrack for the Freescale Cup intelligent car racing competition.

HTML | C | C++ | Python | Arduino | Raspberry Pi | EasyVR

Programming Languages

C++	●●●●●
MATLAB	●●●●●
C	●●●●●
Python	●●●●○
HTML	●●●●○
Javascript	●●●○○
CSS	●●●○○

Simulation Tools

- OpenDSS
- GridLAB-D
- Simulink
- MATPOWER
- EnergyPlus
- FPGA Design
- Raspberry Pi
- Arduino Uno

Publications

2019 Gayathri Krishnamoorthy, and Anamika Dubey, “Transmission-Distribution Co-Simulation : Analytical Methods for Iterative Coupling, submitted to IEEE Systems Journal (In revision).

2019 Bedoya, J.C., Liu, C.C., Krishnamoorthy, G. and Dubey, A., 2019. Bilateral Electricity Market in a Distribution System Environment. IEEE Transactions on Smart Grid.

2019 Rabayet Sadnan, Gayathri Krishnamoorthy, and Anamika Dubey “Comparison of different coupling method used in transmission and distribution (T&D) co-simulation’, submitted to IET GT&D.

2019 Krishnamoorthy, Gayathri, and Anamika Dubey. “Iteratively-Coupled Co-simulation Framework for Unbalanced Transmission-Distribution System”, accepted for publication in IEEE PES PowerTech Milano 2019.

2018 Krishnamoorthy, Gayathri, and Anamika Dubey. ”A framework to analyze interactions between transmission and distribution systems.” In 2018 IEEE Power & Energy Society General Meeting (PESGM), pp. 1-5. IEEE, 2018.

Projects

Transmission -Distribution Co-simulation 2017 - 2019

github.com/yvelaga/PSERC-Cosimulation PSERC T-60

Developed a framework for the integrated T&D system analysis to understand the impacts of distribution-connected renewable resources (DERs) on transmission system operations and vice-versa (project funded by PSERC). Also, using the co-simulation framework developed, explored the utility of DER's in transmission system support.

MATLAB | OpenDSS | Python

Website for Distribution lab at WSU 2019

DOPAL

Designed a professional website for my research group at WSU.

HTML | bootstrap | CSS | jquery

Real Time Cyber-Physical System Testbed Development 2017

Course Project

Developed a real time cyber-physical system testbed communicating with each other through MatrikonOPC software to analyze the impacts of a real-time cyberattack on the power grid. Designed an Intrusion Detection System (IDS) in substations using SNORT software to provide additional security along with firewalls.

Matrikon OPC | MATLAB | Simulink | C++

Multi-core Processor Design 2016

github.com/adititihuse/Branch-Target-Buffer-Simulation

Development of an advanced multi-core processor design in C++ that handles functional units looping instructions for static and dynamic branch predictions

C++

Honors and Awards

2019	**Awardee** Graduate & Professional Student Association Scholarship Recipient for the Research Exposition at WSU
2018	**2nd Place** Best Student Paper Award at North American Power Symposium
2015	**Awardee** Completed course in CCNA Exploration : Network Fundamentals
2015	**2nd Place** Texas Instruments India Analog Maker Competition
2014	**Awardee** Level 1 Cambridge English Language Assessment in Business English
2011-2012	**Scholarship** 100 % fee concession in high school

Professional Activities

Present	Serving as a reviewer for IEEE Power and Energy Society Conferences, and Institution of Engineering and Technology Journals
2019	Authored a Ph.D. Research Proposal to Pacific Northwest National Laboratory
2018	Paper Presentation- IEEE Power and Energy Society General Meeting
2017	Poster Presentation - Power Systems Engineering Research Center
2015	Attended Internet of Things with Arduino workshop for real-time applications at AltSense
2015	Trained and participated in FreescaleCup Intelligent Car Racing Competition

References

Anamika Dubey
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